

Microbes in UC Davis

(Specifically outside the Katherine Esau Hall)

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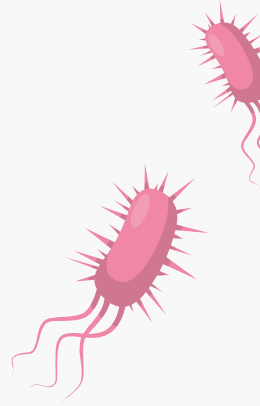
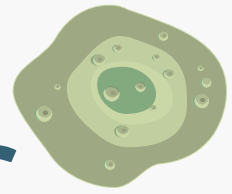
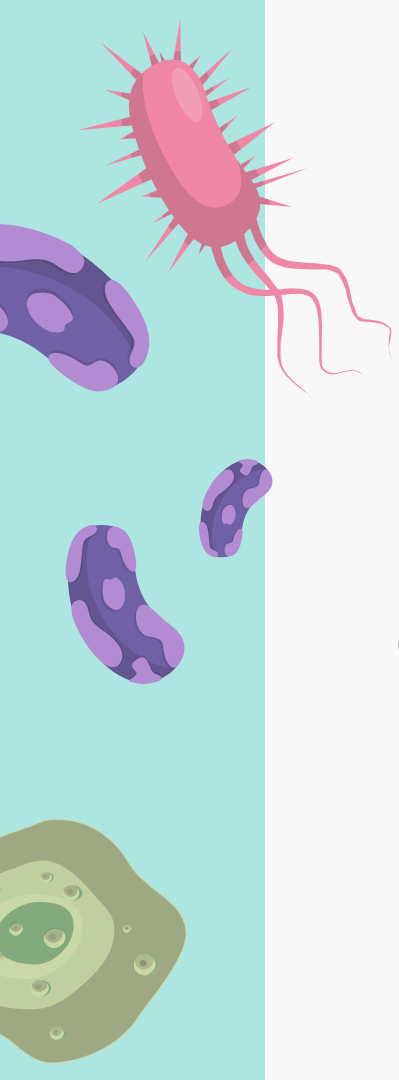


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Introducing the
Project



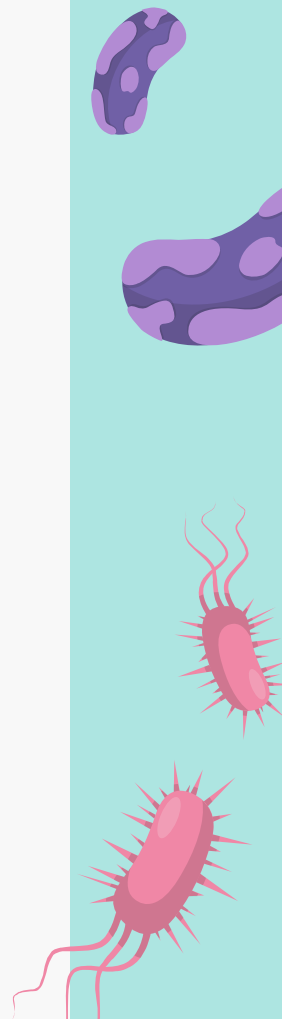
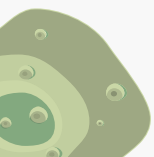
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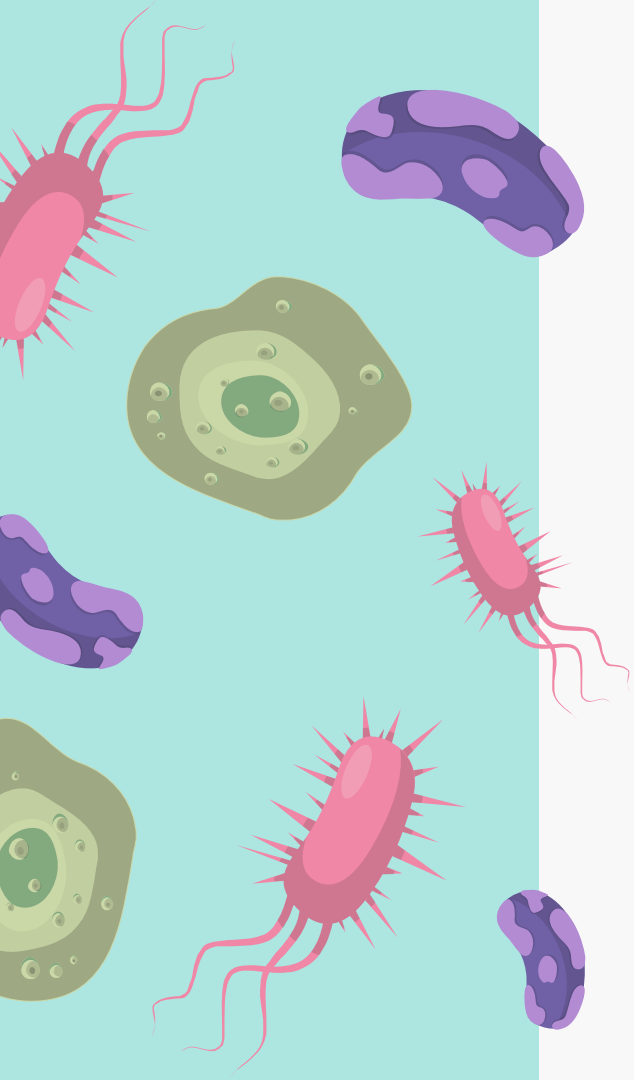
Big Takeaways



03

App Showcase

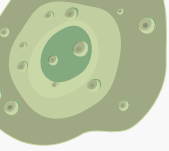




01

Microbes In our Environment





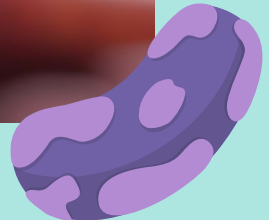
Experiment and Data provided by Dr. Markum's MIC 104L Course

GERMS ARE EVERYWHERE

But which germs are really out there?

That is the question that we tried to answer over the years in the MIC 104L through the Microbes in the Environment Lab procedure.

I was lucky enough to not only participate in the experiment but also was lucky enough to get the previous 2 years worth of data,



Setup and Context

When?

The class is divided into multiple morning sections from 9am-11:50am and multiple afternoon sections from 1:10pm-4pm

What?

Each class section is filled with 24 students, each of which is provided with either 1 SDA + Ampicillin plate or 1 NA + Cycloheximide plate.

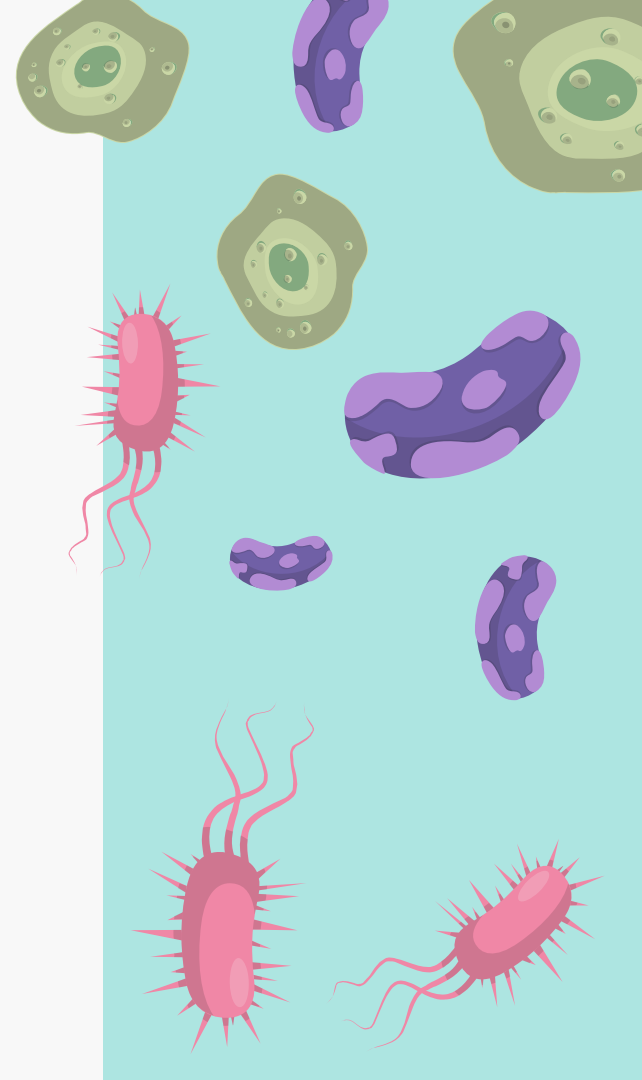
12 students are instructed to go outside and expose their plates to the air and 12 stay inside to expose their plates in the lab.

In either outside or inside groups, 4 students are exposing their plates for 1 minute, 4 for 10 minutes and 4 for 30 minutes.

Why?

SDA plates offer Dextrose sugar as the main carbon source with Ampicillin as the selective factor

NA plates offer Glucose sugar as the main carbon source with Cycloheximide as the selective factor



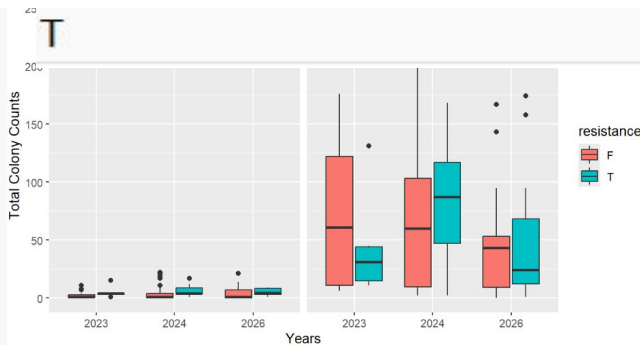
02 Big Takeaways

Resistance

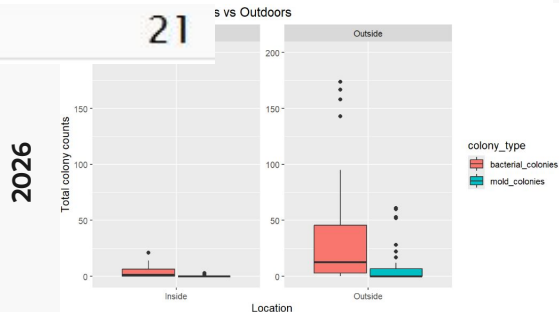
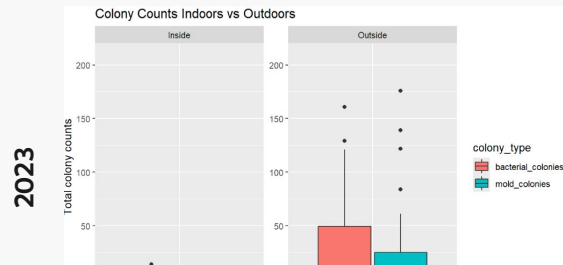
Antibiotic/Antifungal Resistance over the years



year <chr>	location <chr>	min_total_colonies <dbl>	mean_total_colonies <dbl>	median_total_colonies <dbl>	max_total_colonies <dbl>
2023	Inside	0	3.041667	2.0	15
2023	Outside	6	63.625000	44.0	176
2024	Inside	0	4.291667	2.0	22
2024	Outside	2	82.826923	77.5	246
2026	Inside	0	4.250000	3.0	21
2026	Outside	0	44.861111	30.5	174



Prevalence





Our Shiny App demo

